REVISED TECHNICAL DISCLOSURE

Log-Prime Encoded Optical Multiplexing and Aggregate Readout

Corrected receiver physics, decoding model, optical-logic qualification, and experimental validation plan

Document field	Value
Author / inventor	Sebastian Shepherd
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Purpose	Public technical description, reproducibility, and prior-art-style disclosure
Suggested licences	CC BY-SA 4.0 paper; AGPL-3.0-or-later software; CERN-OHL-S-2.0 hardware files

Important status: This document describes a mathematical coding method and a proposed optical implementation. The recorded 12-lane result is a software simulation, not a physically calibrated optical measurement. Novelty and patentability have not been legally determined.

Prepared as a corrected replacement for Log_Prime_Optical_Encoding.pdf

Revision Summary

This revision preserves the core log-prime concept while separating the exact mathematics, the calibrated receiver model, the simulated demonstration, and the still-unverified hardware claims.

Correction	Revised treatment
Power/current physics	Derives P/I explicitly from photon flux and wavelength-dependent quantum efficiency. P/I is a mean photon-energy proxy only after calibration or when effective quantum efficiency is sufficiently constant.
Equal-contribution definition	Defines equality as equal calibrated charge contribution q eta_i Phi_i, not merely equal optical power.
Affine offset B	Shows that the prime-product reconstruction uses the normalized mean (Cbar-B)/A. The offset cancels only after this normalization.
Pixel snapping	Distinguishes ideal target frequencies from actual pixel-centre frequencies. Exact factorisation requires per-lane calibration or decoding against a measured codebook.
NOR stage	States that passive masking alone cannot produce a NOR high output when both inputs are dark. A probe/bias field, nonlinear gate, gain element, or equivalent active optical mechanism is required.
ADC precision	Removes the claim that 3.3 V / 4096 is the practical state spacing. Real robustness is set by the minimum distance between actual valid calibrated codepoints.
Empty output	Defines a separate dark-state decision because P/I is undefined when I approaches zero.
Weighted mode	Requires non-negative integer exponents for direct prime-factor recovery. Other weights require a finite calibrated codebook.
Software appendices	Treats source listings as repository artefacts instead of typesetting duplicated code into the core disclosure.

Abstract

This disclosure describes an optical channel code in which an intended calibrated lane coordinate is generated from a distinct prime number p_i according to C_i = A log(p_i) + B. For a set of active lanes with known equal calibrated contributions, the normalized aggregate mean recovers the sum of log-prime coordinates. Multiplication by the active count and exponentiation produces the product of the active primes, whose factorisation identifies the active lane set under ideal arithmetic. A weighted extension permits non-negative integer lane weights to appear as prime exponents.

A physically defensible aggregate receiver is modelled using total optical power P and total photocurrent I. Their ratio has units of volts. If channel-dependent split loss and quantum efficiency are calibrated, P/I can be transformed into a photon-flux-weighted mean photon energy or mean optical frequency. The method therefore requires two aggregate observables, or equivalent calibrated observables: a total-weight coordinate and a mean-code coordinate. It does not assume that an uncalibrated photodiode voltage directly equals average photon energy.

The accompanying 12-lane result validates the digital mapping, mapped-lane Boolean convention, and ideal prime-product decoder in software. It does not yet demonstrate a physical optical NOR gate or a calibrated aggregate receiver. A practical system should decode by nearest valid codeword using measured per-lane calibration data, uncertainty estimates, and dark-state handling, with exact prime reconstruction retained as an ideal mathematical interpretation.

1. Scope and contribution

The contribution disclosed here is the use of prime-derived logarithmic lane coordinates together with aggregate mean-and-weight recovery. Established components such as wavelength-division multiplexing, gratings, lenses, spatial light modulators, optical power meters, photodiodes, interferometers, nonlinear optical gates, and ADCs are not claimed individually.

The disclosure covers free-space and integrated implementations, provided that the physical receiver produces calibrated observables equivalent to total lane weight and weighted mean code coordinate.

2. Definitions and notation

Each permitted lane i is assigned a distinct prime p_i. The logarithm base may be natural, base 2, or another fixed base, provided the same base and scale are used consistently.

Symbol	Definition
p_i	Distinct prime assigned to lane i
C_i	Ideal lane code coordinate A log(p_i) + B
Phi_i	Photon flux in lane i at the relevant receiver reference plane
eta_i	Effective photon-to-collected-charge efficiency, including calibrated path effects
w_i	Calibrated non-negative lane contribution or discrete integer weight
W	Total calibrated weight, sum of w_i
P	Aggregate optical power
I	Aggregate photocurrent

M_C	Weighted mean lane code coordinate
S	Active lane set

3. Ideal log-prime decoding theory

3.1 Equal calibrated contributions

$$C_i = A \log(p_i) + B$$

$$M_C = (1/N) \sum_{i \in S} C_i$$

Normalize the measured mean before reconstructing the product:

$$M_{log} = (M_C - B) / A = (1/N) \sum_{i \in S} \log(p_i)$$

$$\exp(N M_{log}) = \prod_{i \in S} p_i$$

Unique prime factorisation identifies S in exact arithmetic when each lane occurs at most once, N is known, and the recovered product is sufficiently accurate.

3.2 Integer-weighted contributions

$$M_C = [\sum_i w_i C_i] / W, \quad W = \sum_i w_i$$

$$M_{log} = (M_C - B) / A$$

$$\exp(W M_{log}) = \prod_i p_i^{(w_i)}$$

Direct interpretation of factor exponents requires w_i to be non-negative integers. Rational, signed, or continuous weights do not generally map to ordinary integer prime exponents. Such variants require rescaling to integers, additional sign channels, or nearest-codeword decoding over a finite codebook.

Base consistency: If C_i uses log2, reconstruction is $2^{(W M_{log})}$. If C_i uses ln, reconstruction is $\exp(W M_{log})$. Mixing bases introduces a systematic decoding error.

4. Corrected aggregate receiver physics

Let Phi_i denote photon flux after defining a common receiver reference plane. Let a_i account for the fraction of lane i reaching the optical-power path, and let b_i eta_i account for the fraction and effective charge-collection efficiency in the photocurrent path.

$$P = \sum_i a_i \Phi_i h \nu_i$$

$$I = q \sum_i b_i \eta_i \Phi_i$$

$$P/I = (h/q) \left[\sum_i a_i \Phi_i \nu_i\right] / \left[\sum_i b_i \eta_i \Phi_i\right]$$

Therefore P/I is not automatically the mean photon energy divided by charge. It becomes proportional to a photon-flux-weighted mean frequency only when the channel response factors are equal or when calibration transforms the observations to an equivalent common response:

$$P/I = (h / q \eta_{eff}) \left[\sum_i \Phi_i \nu_i\right] / \left[\sum_i \Phi_i\right] \quad \text{(calibrated special case)}$$

$$\text{mean photon energy} = h \cdot \text{mean}(\nu)$$

The system should therefore use a calibration matrix or per-lane correction coefficients. A practical decoder may operate directly on calibrated P and I coordinates rather than forcing the hardware output into an exact voltage law.

4.1 What equal contribution means

Equal optical power per lane does not imply equal photon flux or equal photocurrent. For equal optical power, photon flux scales approximately as 1/nu. Constant-contribution decoding should instead target one of the following explicitly:

- equal photon flux at the receiver reference plane
- equal calibrated photocurrent contribution q b_i eta_i Phi_i
- equal fitted decoder weight obtained from single-lane calibration

The recommended engineering definition is equal calibrated photocurrent contribution, because the active count N is then inferable from I after dark-current subtraction and linearity calibration.

4.2 Dark state

When no lane is active, both P and I ideally approach zero and P/I is undefined or noise dominated. The receiver must first apply a dark-state test. Only if I and P exceed calibrated thresholds should it compute the ratio or mean-code coordinate.

5. Target coordinates, physical wavelengths, and calibration

The prime law generates target frequencies. A grating/lens/LCD implementation then snaps each target to a physical pixel column. The actual centre wavelength of that pixel generally differs from the target. Consequently, the realised lane coordinates are not exactly A log(p_i) + B unless a tunable source, calibrated filter, or correction stage enforces those coordinates.

Coordinate	Use
Target frequency f_i*	Ideal prime-derived design coordinate
Target wavelength lambda_i*	c/f_i*
Selected pixel x_i	Integer LCD column
Measured centre frequency f_i,cal	Experimentally calibrated lane centre
Measured response vectors	Single-lane P, I, transmission, noise, and drift statistics

Two valid decoder strategies follow:

- Idealised factor decoder: correct each measured aggregate into the target-coordinate model, then apply prime-product reconstruction.
- Calibrated codebook decoder: predict or measure the aggregate codepoint of each permitted state and select the most likely state. This is more robust when snapping, detector response, or optical losses perturb the exact prime law.

Recommended wording: Use "calculated target mapping" for modelled positions and "measured calibrated mapping" only after spectrometer or narrowband-source calibration.

6. Qualification of the optical NOR stage

The software implements the Boolean rule OUT[x] = mapped(x) AND NOT(INPUT1[x] OR INPUT2[x]). This is a valid lane-wise logical definition. It is not by itself a physical optical NOR gate.

A passive mask can only attenuate incident light. It cannot create a bright output solely because two signal inputs are absent. A physical NOR embodiment therefore requires an optical bias or probe beam and a mechanism by which either input suppresses, diverts, phase-cancels, or switches that probe. Candidate mechanisms include interferometric cancellation, saturable or nonlinear absorption, cross-phase modulation, resonant switching, magneto-optic switching, or another externally powered optical gate.

Status	Claim that can be supported
Current software	Mapped-lane Boolean NOR simulation
Passive LCD masks only	Selection or blocking of pre-existing illuminated lanes
Bias/probe plus validated interaction	Potential physical optical NOR, subject to measured truth table
Future magneto-photonic version	Optically controlled magnetic state modulating a probe without intermediate electronic detection

A physical NOR claim should be made only after measuring all four input combinations, output contrast, threshold, extinction ratio, latency, energy per operation, repeatability, and cascadability.

7. Robust decoding procedure

Exact integer reconstruction is useful for theory and noiseless software tests, but a hardware decoder should use uncertainty-aware nearest-codeword or maximum-likelihood decoding.

1. Acquire synchronous P and I measurements, or equivalent aggregate observables.

2. Subtract dark current, background power, offsets, and drift estimates.

3. Apply path split, responsivity, efficiency, gain, and temperature calibration.

4. Test the dark state before forming P/I.

5. Estimate total weight W from the calibrated total-contribution coordinate.

6. Form a normalized mean-code estimate, including removal of B and division by A.

7. Generate candidate states consistent with W and permitted lane/weight constraints.

8. Compare the measured observation vector with calibrated candidate codepoints using the full covariance or an engineering tolerance model.

9. Return the best candidate only if its confidence and margin over the second-best candidate exceed defined thresholds; otherwise return undecodable or request a repeated measurement.

$$\text{decoded state} = \arg\min_s \; (y - \mu_s)^T \Sigma_s^{-1} (y - \mu_s)$$

Here y may contain [P, I], [I, P/I], repeated samples, temperature, or other calibrated observables. μ_s and Σ_s are the expected observation and uncertainty for state s.

8. Precision and code-distance analysis

The original 3.3 V / 4096 estimate assumes 4096 uniformly spaced scalar states. Log-prime mean states are not uniformly spaced, and multiple active counts create a two-dimensional code geometry. The estimate must not be presented as a hardware requirement.

$$\text{Delta_min} = \min_{s \neq t} \| \text{whiten}(\mu_s - \mu_t) \|$$

The relevant analysis enumerates all permitted states, computes their calibrated observation vectors, and finds the nearest neighbours. For a 12-lane binary codebook, 4096 states are small enough to enumerate directly. The report should include:

- minimum and percentile codepoint separations
- which state pairs are nearest neighbours
- noise and drift assumptions
- Monte Carlo symbol and lane error rates
- rejection rate under confidence thresholds
- performance versus number of active lanes
- sensitivity to wavelength shift, power imbalance, responsivity error, and ADC quantisation

Code correction: Remove compute_readout_accuracy_for_effective_bits() as a physical claim. Replace it with codebook enumeration and measured/simulated nearest-neighbour distance analysis.

9. Experimental architecture

source -> collimation -> spectral dispersion -> calibrated mask/modulator -> optional physical logic gate -> aggregate receiver

For the two-path receiver, use a characterised splitter and synchronised measurements:

Path A: aggregate optical power meter -> P

Path B: linear photodiode transimpedance/current receiver -> I

The use of a broadband LED is suitable for spectral-mask demonstrations if individual lane bands are sufficiently separated and calibrated. It does not by itself establish coherent interference or a nonlinear optical logic function. Coherent phase gates generally require an appropriate coherent source and phase-stable paths.

9.1 Minimum calibration record

Measurement	Required record
Lane centre and bandwidth	Measured spectrum for every open lane
Single-lane P and I	Mean, standard deviation, and linearity versus source level
Dark/background	P_dark, I_dark, temperature, integration time
Cross-talk	Response in all other lanes when one lane is opened
Split-path calibration	Wavelength dependence of power/current path ratio
Codebook	Predicted and measured points for permitted combinations
Logic truth table	All logical inputs, output contrast, threshold, and repeatability
Timing	Acquisition window, gate latency, and drift timescale

10. Recorded software demonstration

The existing software run is retained as evidence of the ideal mathematical pipeline. It should be labelled simulation throughout.

Field	Recorded value
Model	12 generated lanes; mapped-lane NOR simulation
Wavelength design window	495 to 740 nm
Mapped LCD columns	0, 4, 6, 12, 17, 20, 28, 33, 49, 63, 87, 110
Input 1 active channels	1, 5, 12
Input 2 active channels	1, 2, 6, 8, 10, 11
Simulated NOR output	Channels 3, 4, 7, 9
Ideal decoder result	Exact match in noiseless arithmetic
Physical measurement status	Not established by this run

The result proves that the software can generate a mapping, apply the stated Boolean rule, calculate its ideal aggregate code, and recover the corresponding subset. It does not prove optical accuracy, physical NOR behaviour, detector precision, or robustness to noise.

11. Software corrections required

Component	Required change
main.py	Delete or rename the uniform 2^b voltage-spacing function. Report codebook separations instead.
nor_voltage_decoder.py	Separate simulated-code mode from physical-receiver mode. In physical mode accept calibrated P and I plus uncertainty, not only an invented V_OFFSET + K mean-log voltage.
Wavelength_LCD_Encoder.py	Preserve both ideal target and calibrated actual coordinates. Do not use pixel snapping error alone as proof of spectral separability.
LCD codec	Restore the intended physical LCD height when amplitude encoding is tested. Current historical listing uses LCD_HEIGHT = 1, so it cannot demonstrate multi-level vertical weighting.
Optical_LCD_Codec.py	Correct the duplicated random_input module header and ensure the file actually contains the intended codec implementation.
Product decoder	Use base-consistent exponentiation and reject products containing repeated factors in binary mode.
Hardware decoder	Add dark-state handling, confidence/rejection logic, calibration versioning, and repeated-sample support.
Repository	Keep full source code in version-controlled files and cite a release hash/DOI rather than embedding duplicate listings in the paper.

12. Supported claims and limitations

12.1 Claims supported by the present record

- A formal log-prime lane-coordinate and aggregate decoding construction.
- A software-generated 12-lane wavelength-to-pixel design map.
- A mapped-lane Boolean NOR simulation.
- Exact recovery of one noiseless simulated output subset.
- A proposed two-observable aggregate receiver and calibration method.

12.2 Claims not yet supported

- Physical operation of an optical NOR gate.
- Recovery of 4096 physical states across a 3.3 V range.
- A specified hardware voltage accuracy requirement based on uniform spacing.
- Physical proof that P/I tracks the intended log-prime mean without calibration.
- High-speed, low-error, or commercially scalable operation.
- Novelty over all prior art or patentability.

12.3 Fundamental engineering limits

- finite optical bandwidth and channel overlap
- photodetector shot noise, thermal noise, nonlinearity, and saturation
- wavelength-dependent source, mask, splitter, grating, and detector response
- growth of codebook density as lane count and weight range increase
- exponential growth in unrestricted state count even if detector count remains small
- factor-product numerical sensitivity and the need for rejection rather than forced decoding
- loss, fan-out, and cascadability limits for detectorless intermediate optical or magneto-photonic logic

13. Detectorless intermediate processing extension

The aggregate decoder is principally an end-of-pipeline readout method. It does not require intermediate detection. Multiple wavelength, mode, or polarisation lanes may remain multiplexed through passive, nonlinear, or magneto-optic transformations, followed by aggregate or demultiplexed detection only at the output boundary.

A future magneto-photonic embodiment could use optically switched magnetic states for nonvolatile control, routing, phase shifts, or probe modulation. Such elements should be described as stateful optical transformations rather than as direct sensors of the instantaneous optical magnetic field. The key system tests would be gain or restoration, fan-out, isolation, cross-talk, switching energy, and cascadability.

14. Validation plan

Phase A - single-lane calibration

- Measure spectrum and centre frequency of each selected pixel column.
- Measure P and I for each lane over source power and temperature.
- Fit channel response coefficients and uncertainty.

Phase B - subset codebook

- Measure all 4096 binary combinations if practical, or a statistically designed subset followed by prediction checks.
- Calculate nearest neighbours and confusion matrix.
- Compare exact factor decoding against nearest-codeword decoding.

Phase C - physical logic

- Implement a real probe/bias NOR mechanism.
- Measure four-state truth table, extinction ratio, latency, and repeatability.
- Cascade at least two stages without electronic detection to test restoration and loss.

Phase D - scale and multiplexing

- Increase lane count while tracking minimum code separation.
- Test wavelength, polarisation, and mode multiplexing.
- Evaluate aggregate detector count, calibration time, computational decoder cost, and error rate.

15. Publication and licensing statement

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Recommended reciprocal licensing remains: CC BY-SA 4.0 for the paper and diagrams, AGPL-3.0-or-later for software, and CERN-OHL-S-2.0 for hardware design files. Repository releases should include CITATION.cff, source files, generated maps, calibration data, raw measurements, scripts used to produce figures, and a DOI-bearing archival release.

Appendix A - Mathematical conditions for unique ideal recovery

- Every lane uses a distinct prime.
- The decoder knows A, B, and the logarithm base.
- The total active count N or total integer weight W is independently known or recoverable.
- Binary lanes occur at most once, or integer-weighted lanes use allowed non-negative exponents.
- The normalized mean is accurate enough that a unique permitted state remains after tolerance is applied.
- The physical lane coordinate is corrected to the ideal prime-derived coordinate, or decoding uses a calibrated state codebook.

Appendix B - Recommended terminology

Avoid	Use instead
measured voltage, for simulated value	simulated code value or V-equivalent
actual optical NOR	simulated mapped-lane NOR, unless physically measured
photodiode voltage equals mean photon energy	calibrated aggregate P/I is transformed into a mean-energy estimate under stated assumptions
4096 equally spaced states	4096 permitted states with measured or simulated non-uniform codepoint geometry
physical wavelength map, for calculated geometry	calculated target map
exact hardware tolerance	ideal integer-reconstruction tolerance or measured decoding margin

Appendix C - Repository artefacts

The full historical source listings should remain in the associated repository rather than being duplicated in this paper. The archival release should identify each file, version, checksum, execution command, dependency version, and whether outputs are simulated or measured.